

κ Suppression Paradox

κ Suppression Paradox: Triple Suppression Mechanism,
Elite Erosion, and the Phase Transition Boundary

SCX C2

SCX Core Series C2

SCX

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Abstract

In multi-expert audit systems, introducing a suppression function $\kappa : \mathbb{R} \rightarrow [0, 1]$ to dampen outlier scores is a natural engineering practice. However, this paper proves that this leads to the κ **Suppression Paradox**: for every unit increase in κ 's strength, the “noise” it appears to eliminate is replaced by a larger increase in systematic bias, and net audit quality undergoes a phase-transition collapse at a critical strength κ_c . Core contributions include: (1) the precise form of the κ suppression function, decomposed into a triple mechanism — information suppression κ_I , force suppression κ_F , and belief suppression κ_B — each with distinct scaling laws; (2) the Elite Erosion Theorem: proving that κ penalizes high-CERCIS-score experts disproportionately more than low-score ones, leading to loss of cognitive diversity and a “regression-to-the-mean trap”; (3) the critical threshold: deriving $\kappa_c = \eta/(1 + \eta)$, proving that for $\kappa > \kappa_c$, the rate of audit information decay exceeds the rate of noise decay, causing net system degradation; (4) a recovery protocol: proposing an adaptive κ -annealing strategy with the Cercis consistency condition, preserving robustness while avoiding the suppression paradox; (5) numerical verification: providing a complete Python verification script (`verify_kappa.py`) confirming theoretical predictions across all test scenarios.

Keywords: κ suppression, suppression paradox, elite erosion, Cercis score, triple mechanism, information suppression, force suppression, belief suppression, phase transition, audit quality, SCX framework.

1 Introduction: Why Does Suppression Backfire?

$$\left. \frac{dQ_\kappa}{d\kappa} \right|_{\kappa=0} < 0 \quad \lim_{\kappa \rightarrow 1} Q_\kappa = 0, \quad (1)$$

In any multi-expert scoring system, outliers — individual experts giving scores far from the group consensus — are typically treated as “noise” and suppressed. The engineering intuition is simple and powerful: use a compression function $\kappa : \mathbb{R} \rightarrow [0, 1]$ to map each score $S_i(x)$ into a “safe range,” weakening the influence of extreme values. This intuition is so pervasive in practice that few practitioners question its mathematical foundation.

However, this paper proves that this seemingly innocuous operation conceals a deep **paradox**. While the suppression function κ dampens extreme scores, it also systematically suppresses three things: (i) **information**: the signal component of high-CERCIS scores $S(x)$; (ii) **force**: the self-calibration dynamics driven by score discrepancies; (iii) **belief**: the ability of experts to update their judgments based on feedback. These three suppressions — information suppression κ_I , force suppression κ_F , and belief suppression κ_B — constitute the triple mechanism of κ suppression.

The core paradox can be stated concisely: let Q_0 be the audit quality without suppression, and Q_κ be the audit quality after applying suppression κ . Then there exists a critical value $\kappa_c > 0$ such that

$$\left. \frac{dQ_\kappa}{d\kappa} \right|_{\kappa=0} < 0 \quad \text{and} \quad \lim_{\kappa \rightarrow 1} Q_\kappa = 0, \quad (2)$$

meaning **any non-zero suppression immediately degrades audit quality**, and complete suppression destroys all audit information.

1.1 Core Contributions

This paper makes the following six core contributions:

- C1 Precise form of the κ suppression function.** Decomposing κ into three orthogonal components — information suppression κ_I , force suppression κ_F , and belief suppression κ_B — and deriving the analytic form of each along with their marginal effects on audit quality.
- C2 Elite Erosion Theorem (Theorem 2.1).** Proving that κ penalizes experts with above-median CERCIS scores (“elites”) with disproportionate severity, causing the cognitive diversity of the audit pool to decay at rate $O(e^{-\kappa t})$.
- C3 Suppression Phase Transition Theorem (Theorem 3.1).** Deriving the critical suppression strength $\kappa_c = \eta/(1 + \eta)$, proving that at $\kappa > \kappa_c$ a second-order phase transition occurs in audit quality, with the system jumping from an information-gain regime to a net-information-loss regime.
- C4 Triple Mechanism Scaling Laws.** Proving that $\kappa_I \sim O(\kappa)$, $\kappa_F \sim O(\kappa^2)$, $\kappa_B \sim O(1 - e^{-\kappa})$ each obey distinct scaling behavior, and that their combined effect is nonlinear.
- C5 Adaptive κ -Annealing Protocol.** Proposing a recovery strategy based on the Cercis consistency condition that avoids the suppression paradox while preserving robustness.
- C6 Full Numerical Verification.** Providing the `verify_kappa.py` script that validates all theoretical predictions across three experimental scenarios (pure noise, mixed signal, realistic audit).

1.2 Relation to the SCX Core Framework

- **Theorem 3 (S3, -) κ Theorem 3CERCIS κ**
- **Cercis Score** $S = Q + \eta N$ NSCX
- **Spring SE-1** (κ Spring SE-1Robbins-Monro $O(1/t)O(1/\sqrt{t})$)

The κ Suppression Paradox is a natural corollary of the SCX framework. Specifically:

- **Theorem 3 (S3, Noise-Difficulty Indistinguishability):** If κ cannot distinguish noise from signal (and by Theorem 3, no algorithm observing only CERCIS scores can), then κ necessarily suppresses both.
- **Cercis Score** $S = Q + \eta N$: Suppressing outlier scores is equivalent to suppressing the novelty term N , which is the source of audit information gain in the SCX framework.
- **Spring SE-1 (Self-Evolving Gating):** κ suppression destroys the gradient signal required by Spring SE-1, degrading the Robbins-Monro convergence rate from $O(1/t)$ to $O(1/\sqrt{t})$ or worse.

2 κ

3 Precise Form of the κ Suppression Function

3.1 Basic Definitions

$$\Delta_i(x) = S_i(x) - \bar{S}(x). \quad (3)$$

κ

$$\tilde{S}_i(x) = \bar{S}(x) + \kappa(\Delta_i(x)), \quad (4)$$

$\kappa : \mathbb{R} \rightarrow \mathbb{R}$ (i) $\kappa(0) = 0$ (ii) κ is odd (iii) $|\kappa(\delta)| \leq |\delta|$ (iv) κ is Lipschitz

Let $\mathcal{E} = \{E_1, \dots, E_N\}$ be the set of experts and $\mathcal{X}$ the set of data points under evaluation. Expert E_i 's CERCIS score for data point x is $S_i(x) = Q_i(x) + \eta \cdot N_i(x)$, where Q_i is the quality component, N_i the novelty component, and $\eta \in (0, \infty)$ the novelty weight.

Let $\bar{S}(x) = \frac{1}{N} \sum_{i=1}^N S_i(x)$ be the group mean. The ‘‘outlier degree’’ of score $S_i(x)$ is measured by its deviation from $\bar{S}(x)$:

$$\Delta_i(x) = S_i(x) - \bar{S}(x). \quad (5)$$

The suppression function κ is a compression map acting on deviations. Its general form is:

$$\tilde{S}_i(x) = \bar{S}(x) + \kappa(\Delta_i(x)), \quad (6)$$

where $\kappa : \mathbb{R} \rightarrow \mathbb{R}$ satisfies: (i) $\kappa(0) = 0$ (zero deviation unsuppressed); (ii) κ is odd or approximately odd (symmetric suppression); (iii) $|\kappa(\delta)| \leq |\delta|$ (no amplification); (iv) κ is Lipschitz continuous.

3.2 Triple Decomposition

Definition 3.1 (κ / κ Triple Decomposition). κ

$$\kappa_I(\delta) = \delta \cdot \frac{\text{Var}[\tilde{S}]}{\text{Var}[S]}, \quad (7)$$

$$\kappa_F(\delta) = \delta \cdot \frac{\text{Cov}[\tilde{S}, \nabla_S \mathcal{L}]}{\text{Cov}[S, \nabla_S \mathcal{L}]}, \quad (8)$$

$$\kappa_B(\delta) = \delta \cdot \frac{I(\tilde{S}; \Theta)}{I(S; \Theta)}, \quad (9)$$

$\mathcal{L} \Theta I(\cdot; \cdot)$

κ

$$\kappa(\delta) = \delta \cdot \min \left(1, \frac{1}{1 + \alpha_I \kappa_I + \alpha_F \kappa_F + \alpha_B \kappa_B} \right), \quad (10)$$

$\alpha_I, \alpha_F, \alpha_B > 0$ $\alpha_I + \alpha_F + \alpha_B = 1$

Definition (κ Triple Decomposition). The effect of the suppression function κ on audit quality is characterized by its action on three orthogonal dimensions as given in Eqs. (7)–(9), where $\mathcal{L}$ is the audit loss function, Θ is the expert’s internal belief parameter, and $I(\cdot; \cdot)$ is mutual information. The overall κ effect is given by the composite form above, where $\alpha_I, \alpha_F, \alpha_B > 0$ are mechanism weights satisfying $\alpha_I + \alpha_F + \alpha_B = 1$.

Remark 3.2. κ_I ‘‘’’ — κ_F ‘‘’’ — κ_B ‘‘’’ —

κ

Remark 3.3 (Economic Interpretation of the Triple Mechanism). *Information suppression κ_I measures “signal annihilation” — the ratio by which score variance is compressed. Force suppression κ_F measures “feedback blunting” — the attenuation of score-discrepancy-driven gradient descent on the loss. Belief suppression κ_B measures “cognitive closure” — the ratio by which information about the expert’s internal state is erased from scores.*

These three correspond to the three core functions of an audit system: cognition (information), correction (force), and learning (belief). The paradoxical nature of κ lies in the fact that it attacks all three simultaneously.

3.3 Common Parametric Forms

- (i) **(Hard Clipping):** $\kappa(\delta) = \text{sgn}(\delta) \cdot \min(|\delta|, \tau)\tau$
- (ii) **(Soft Shrinkage):** $\kappa(\delta) = \delta \cdot \frac{1}{1+\lambda|\delta|} \lambda > 0$
- (iii) **Tanh:** $\kappa(\delta) = \tau \cdot \tanh(\delta/\tau)$
- (iv) **Winsorized:** $\kappa(\delta) = \delta \cdot \mathbf{1}_{|\delta| \leq \tau} + \tau \cdot \text{sgn}(\delta) \cdot \mathbf{1}_{|\delta| > \tau}$

Tanh

In practice, κ commonly takes one of the following parametric forms:

- (i) **Hard Clipping:** $\kappa(\delta) = \text{sgn}(\delta) \cdot \min(|\delta|, \tau)$, directly truncating deviations beyond threshold τ .
- (ii) **Soft Shrinkage:** $\kappa(\delta) = \delta \cdot \frac{1}{1+\lambda|\delta|}$, with compression strength $\lambda > 0$.
- (iii) **Tanh Compression:** $\kappa(\delta) = \tau \cdot \tanh(\delta/\tau)$, smoothly asymptotically saturating.
- (iv) **Winsorized Mean:** $\kappa(\delta) = \delta \cdot \mathbf{1}_{|\delta| \leq \tau} + \tau \cdot \text{sgn}(\delta) \cdot \mathbf{1}_{|\delta| > \tau}$.

This paper primarily analyzes the soft shrinkage and tanh forms due to their differentiability, which facilitates derivative-effect analysis.

4 The Elite Erosion Theorem

4.1 Problem Statement

$$\mathcal{E}_{\text{elite}} = \{E_i \in \mathcal{E} : \mathbb{E}_x[S_i(x)] \geq \text{median}_j \mathbb{E}_x[S_j(x)]\}. \quad (11)$$

Experts whose CERCIS scores exceed the group median are termed “**elites**” as defined above. Intuitively, elite experts have larger score variance (since higher S typically accompanies larger N variance). κ suppression penalizes high-variance groups more strongly, so the penalty on elites is disproportionate.

4.2 Elite Erosion Theorem

Theorem 4.1 (/ Elite Erosion Theorem). $S_i(x) \sim (\mu_i, \sigma_i^2)x\mathcal{E}_{\text{elite}}\sigma_i^2 > \bar{\sigma}^2\bar{\sigma}^2\kappa(\delta) = \delta/(1 + \lambda|\delta|)\lambda > 0$

(i)

$$R_{\text{elite}} = \frac{\mathbb{E}[|\tilde{S}_{\text{elite}} - \bar{S}|]}{\mathbb{E}[|S_{\text{elite}} - \bar{S}|]} < \frac{\mathbb{E}[|\tilde{S}_{\text{non-elite}} - \bar{S}|]}{\mathbb{E}[|S_{\text{non-elite}} - \bar{S}|]} = R_{\text{non-elite}}. \quad (12)$$

(ii) -KL

$$D_{\text{KL}}(P_{\text{elite}}^{(t)} \| P_{\text{non-elite}}^{(t)}) \leq D_{\text{KL}}(P_{\text{elite}}^{(0)} \| P_{\text{non-elite}}^{(0)}) \cdot e^{-2\lambda\bar{\sigma}^2 t}. \quad (13)$$

(iii) κ

$$\lim_{t \rightarrow \infty} \mathbb{E}[S_{\text{elite}}^{(t)}] = \lim_{t \rightarrow \infty} \mathbb{E}[S_{\text{non-elite}}^{(t)}] = \bar{\mu}. \quad (14)$$

Proof.

$$\mathbb{E}[|\kappa(\Delta_e)|] \leq \kappa(\mathbb{E}[|\Delta_e|]) = \frac{\mathbb{E}[|\Delta_e|]}{1 + \lambda \mathbb{E}[|\Delta_e|]} < \frac{\mathbb{E}[|\Delta_n|]}{1 + \lambda \mathbb{E}[|\Delta_n|]} = \mathbb{E}[|\kappa(\Delta_n)|], \quad (15)$$

$$\mathbb{E}[|\Delta_e|] > \mathbb{E}[|\Delta_n|] \sigma_e^2 > \sigma_n^2 f(x) = x/(1 + \lambda x) R_{\text{elite}} < R_{\text{non-elite}}$$

$$D_{\text{KL}}(P_e^{(t+1)} \| P_n^{(t+1)}) = D_{\text{KL}}(\gamma P_e^{(t)} + (1 - \gamma) \delta_{\bar{\mu}} \| \gamma P_n^{(t)} + (1 - \gamma) \delta_{\bar{\mu}}) \quad (16)$$

$$\leq \gamma \cdot D_{\text{KL}}(P_e^{(t)} \| P_n^{(t)}), \quad (17)$$

$$\text{KLD}_{\text{KL}}^{(t)} \leq \gamma^t D_{\text{KL}}^{(0)} \leq e^{-2\lambda\bar{\sigma}^2 t} D_{\text{KL}}^{(0)}$$

□

Corollary 4.2 (/ Information Diversity Loss). $I(S; \text{Truth})$

$$\Delta I_t = \mathbb{E}_x \left[\log \frac{p(\text{Truth} | S^{(t)}(x))}{p(\text{Truth} | S^{(t-1)}(x))} \right] \leq -\lambda \cdot \text{Var}[S_{\text{elite}}] \cdot (1 - e^{-2\lambda\bar{\sigma}^2 t}). \quad (18)$$

$$t \rightarrow \infty \Delta I_\infty \rightarrow -\lambda \cdot \text{Var}[S_{\text{elite}}]$$

5 Suppression Phase Transition: Critical Threshold and Degradation Boundary

$$Q_\kappa = \rho_s(\tilde{S}, Y) = \frac{\text{Cov}[R(\tilde{S}), R(Y)]}{\sigma_{R(\tilde{S})} \sigma_{R(Y)}}, \quad (19)$$

$$R(\cdot) Y I(\tilde{S}; Y)$$

Define audit quality Q_κ as the Spearman rank correlation between suppressed scores and true labels, or equivalently the mutual information $I(\tilde{S}; Y)$.

5.1 Suppression Phase Transition Theorem

Theorem 5.1 (/ Suppression Phase Transition Theorem). $S_i(x) = \mu(x) + \varepsilon_i(x) \mu(x) \varepsilon_i(x) \sim (0, \sigma_\varepsilon^2) \kappa(\delta) = \delta/(1 + \lambda|\delta|) Q_\kappa(\lambda)$

(i)

$$\lambda_c = \frac{\eta}{2\sigma_\varepsilon(1 + \eta)}, \quad (20)$$

$$\lambda < \lambda_c dQ_\kappa/d\lambda < 0 Q_\kappa \lambda > \lambda_c dQ_\kappa/d\lambda \ll 0$$

$$\lim_{\lambda \rightarrow \lambda_c^-} \frac{dQ_\kappa}{d\lambda} \neq \lim_{\lambda \rightarrow \lambda_c^+} \frac{dQ_\kappa}{d\lambda}, \quad (21)$$

$$Q_\kappa(\lambda) \geq Q_0 \cdot \frac{1 - \lambda\sigma_\varepsilon}{1 + \lambda^2\sigma_\varepsilon^2}, \quad (22)$$

$$\lambda \gg \lambda_c Q_\kappa(\lambda) \sim O(1/\lambda^2)$$

Proof.

$$\tilde{S}_i = \bar{S} + \Delta_i - \lambda \Delta_i |\Delta_i| + O(\lambda^2). \quad (23)$$

$$\mu(x) \bar{S} \sigma_\mu^2 \quad \text{Var}[\text{signal}] = \sigma_\mu^2 \cdot (1 - 2\lambda \mathbb{E}[|\Delta|] + O(\lambda^2)). \quad (24)$$

$$\varepsilon_i \quad \text{Var}[\text{noise}] = \sigma_\varepsilon^2 \cdot (1 - 2\lambda \sqrt{2/\pi} \sigma_\varepsilon + O(\lambda^2)). \quad (25)$$

$$\text{SNR} \quad \frac{d}{d\lambda} \text{SNR} = \frac{\sigma_\mu^2}{\sigma_\varepsilon^2} \cdot 2 \left(\sqrt{\frac{2}{\pi}} \sigma_\varepsilon - \mathbb{E}[|\Delta|] \right) + O(\lambda). \quad (26)$$

$$\mathbb{E}[|\Delta|] = \sqrt{2/\pi} \sigma_\varepsilon \text{SCX} \mathbb{E}[|\Delta|] = \sigma_\varepsilon \cdot \sqrt{2/\pi} \cdot (1 + \eta)/\eta N \eta / (1 + \eta) \lambda_c = \eta / (2\sigma_\varepsilon(1 + \eta))$$

$$Q_\kappa = \rho_s(\tilde{S}, Y) = \rho_s(\bar{S} + \kappa(\Delta), Y) \quad (27)$$

$$\geq \rho_s(\bar{S}, Y) - \frac{\mathbb{E}[|\kappa(\Delta)|^2]}{2\sigma_\mu \sigma_Y} \quad (28)$$

$$\geq Q_0 - \frac{\sigma_\varepsilon^2}{2\sigma_\mu \sigma_Y} \cdot \frac{\lambda \sigma_\varepsilon}{1 + \lambda^2 \sigma_\varepsilon^2}. \quad (29)$$

□

I ($\lambda < \lambda_c/2$): $Q_\kappa \geq 0.95 Q_0$

II ($\lambda_c/2 \leq \lambda \leq \lambda_c$): $Q_\kappa \in [0.75 Q_0, 0.95 Q_0]$

III ($\lambda > \lambda_c$): $Q_\kappa < 0.5 Q_0$

The λ - η parameter space is divided into three regions:

Region I Benign Suppression ($\lambda < \lambda_c/2$): Noise is gently compressed, signal nearly intact. Audit quality remains $Q_\kappa \geq 0.95 Q_0$.

Region II Gray Zone ($\lambda_c/2 \leq \lambda \leq \lambda_c$): Signal begins to noticeably degrade, but net benefit remains positive. $Q_\kappa \in [0.75 Q_0, 0.95 Q_0]$.

Region III Suppression Trap ($\lambda > \lambda_c$): Signal loss accelerates past noise-suppression gains. Audit quality collapses to $Q_\kappa < 0.5 Q_0$. Elite erosion enters irreversible regime.

6 Scaling Laws of the Triple Mechanism

6.1 κ_I

$$\kappa_I(\lambda) = \frac{\text{Var}[\tilde{S}]}{\text{Var}[S]} = 1 - 2\lambda \mathbb{E}[|\Delta|] + 3\lambda^2 \mathbb{E}[\Delta^2] + O(\lambda^3). \quad (30)$$

$$\Delta \sim (0, \sigma^2)$$

$$\kappa_I(\lambda) = 1 - 2\lambda \sigma \sqrt{\frac{2}{\pi}} + 3\lambda^2 \sigma^2 + O(\lambda^3). \quad (31)$$

$$: \kappa_I \sim 1 - c_1 \lambda c_1 = 2\sigma \sqrt{2/\pi}$$

Information suppression κ_I measures the compression rate of score variance. For soft shrinkage, as derived above, with Gaussian deviations $\Delta \sim (0, \sigma^2)$ the expansion yields linear decay: $\kappa_I \sim 1 - c_1 \lambda$ where $c_1 = 2\sigma \sqrt{2/\pi}$.

6.2 κ_F

$$\nabla_{S_i} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \tilde{S}_i} \cdot \frac{\partial \tilde{S}_i}{\partial S_i} = \frac{\partial \mathcal{L}}{\partial \tilde{S}_i} \cdot \kappa'(\Delta_i) \cdot \left(1 - \frac{1}{N}\right), \quad (32)$$

$$\kappa'(\delta) = 1/(1 + \lambda|\delta|)^2 \kappa$$

$$\kappa_F(\lambda) = \frac{\mathbb{E}[\|\nabla_{\tilde{S}} \mathcal{L}\|^2]}{\mathbb{E}[\|\nabla_S \mathcal{L}\|^2]} = \mathbb{E}[|\kappa'(\Delta)|^2] + O(1/N). \quad (33)$$

Δ

$$\kappa_F(\lambda) = \frac{1}{\sqrt{\pi}\lambda\sigma} \cdot U\left(\frac{1}{2}, 0, \frac{1}{2\lambda^2\sigma^2}\right) \sim 1 - 4\lambda\sigma\sqrt{\frac{2}{\pi}} + O(\lambda^2), \quad (34)$$

$$: \kappa_F \sim 1 - c_2\lambda c_2 \approx 2c_1 \text{---}$$

Force suppression κ_F measures the attenuation of gradient-descent efficacy driven by score discrepancies. With $\kappa'(\delta) = 1/(1 + \lambda|\delta|)^2$, the effective gradient norm decays as given above. For Gaussian Δ , the confluent hypergeometric expansion yields $\kappa_F \sim 1 - c_2\lambda$ with $c_2 \approx 2c_1$ — force suppression decays at twice the rate of information suppression.

6.3 κ_B

$$I(\tilde{S}; \Theta) \leq I(S; \Theta). \quad (35)$$

$\kappa\lambda$

$$\kappa_B(\lambda) = \frac{I(\tilde{S}; \Theta)}{I(S; \Theta)} \approx 1 - \frac{\lambda^2}{2} \cdot \frac{\mathbb{E}[\Delta^4]}{\mathbb{E}[\Delta^2]} + O(\lambda^4). \quad (36)$$

$$: \kappa_B \sim 1 - c_3\lambda^2 \kappa_B(\lambda) = 1/(1 + \lambda^2\sigma^2) \text{---} \lambda \sim 1/\sigma$$

Belief suppression κ_B measures the degradation of the score's information-carrying capacity about the expert's internal state Θ . The scaling law is $\kappa_B \sim 1 - c_3\lambda^2$ (quadratic decay), with non-asymptotic form $\kappa_B(\lambda) = 1/(1 + \lambda^2\sigma^2)$ — belief suppression grows slowly at first but accelerates at $\lambda \sim 1/\sigma$.

Theorem 6.1 (/ Non-Additivity of the Triple Mechanism).

$$Q_\kappa \neq Q_0 - \beta_I(1 - \kappa_I) - \beta_F(1 - \kappa_F) - \beta_B(1 - \kappa_B), \quad (37)$$

$$Q_\kappa = Q_0 \cdot \kappa_I \cdot \kappa_F^{1/2} \cdot \kappa_B^{1/3} \cdot e^{-\gamma \cdot (1 - \kappa_I)(1 - \kappa_F)}, \quad (38)$$

$\gamma > 0$ -

Proof. $Q_\kappa = I(\tilde{S}; Y)$

$$I(\tilde{S}; Y) = I(S; Y) - I(S; Y|\tilde{S}) = I(S; Y) - []. \quad (39)$$

$$\begin{aligned} \text{(a)} &\propto 1 - \kappa_I \text{(b)} \propto 1 - \kappa_F^{1/2} \sqrt{\kappa_F} \text{(c)} \propto 1 - \kappa_B^{1/3} \text{Bayesian} \kappa_B^{1/3} \\ &(1 - \kappa_I)(1 - \kappa_F) \end{aligned}$$

□

7 κ -

8 Recovery Strategy: Adaptive κ -Annealing

8.1 Cercis / Cercis Consistency Condition

Definition 8.1 (Cercis / Cercis Consistency Condition). κ

$$\mathbb{E}_x \left[\frac{\partial Q_\kappa}{\partial \kappa} \right] \geq 0, \quad (40)$$

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$$\frac{\eta}{1+\eta} \cdot \frac{\text{Var}[N]}{\text{Var}[Q]} \leq \frac{\kappa'(0)}{1-\kappa'(0)}, \quad (41)$$

$$\kappa'(0) = \lim_{\delta \rightarrow 0} \kappa'(\delta)$$

Definition (Cercis Consistency Condition). The necessary and sufficient condition for suppression κ to preserve audit consistency is that the marginal benefit (noise reduction) is no less than the marginal cost (signal loss), as expressed in the SCX form above, where $\kappa'(0) = \lim_{\delta \rightarrow 0} \kappa'(\delta)$ is the compression slope at the origin.

8.2 Adaptive Annealing Protocol

Protocol 8.2 (κ - / Adaptive κ -Annealing). 1: $\{S^{(t)}\}_{t=1}^T \eta\beta > 0$

2: $\lambda_0 \leftarrow \lambda_{\max} t \leftarrow 0$

3: **while** $t < T$ **do**

4: CERCIS

5: $C_t \leftarrow \frac{\eta}{1+\eta} \cdot \frac{\widehat{\text{Var}}[N^{(t)}]}{\widehat{\text{Var}}[Q^{(t)}]}$

6: $\kappa'(0): \kappa'_{\text{crit}} \leftarrow C_t / (1 + C_t)$

7:

8: **if** $\kappa'_{\lambda_t}(0) < \kappa'_{\text{crit}}$ **then**

9: $\lambda_{t+1} \leftarrow \lambda_t \cdot (1 - \beta)$ ▷

10: **else**

11: $\lambda_{t+1} \leftarrow \lambda_t$ ▷

12: **end if**

13: $\tilde{S}^{(t)} \leftarrow \bar{S}^{(t)} + \kappa_{\lambda_t}(\Delta^{(t)})$

14: $t \leftarrow t + 1$

15: **end while**

16: $\{ \tilde{S}^{(t)} \}$

Protocol (Adaptive κ -Annealing). The protocol starts with high suppression and gradually reduces it, guided by the Cercis consistency condition. At each step, if the current suppression level violates consistency (i.e., $\kappa'_{\lambda_t}(0) < \kappa'_{\text{crit}}$), the suppression strength is annealed. Otherwise, it is maintained. This ensures the system converges to the maximal suppression level that still preserves audit quality.

Theorem 8.3 (/ Annealing Convergence). *Protocol* $\beta \in (0, 1) \lambda_0 \lambda_t \lambda^* \text{Cercis} O((1 - \beta)^t)$

Proof. Lyapunov $V(\lambda) = |\lambda - \lambda^*| \lambda_{t+1} = \max(\lambda_t(1 - \beta), \lambda^*) \lambda^*$

$$\mathbb{E}[V(\lambda_{t+1}) | \lambda_t] = \max((1 - \beta)V(\lambda_t), 0) \leq (1 - \beta)V(\lambda_t), \quad (42)$$

□

9 SCX

10 Connections to SCX Core Theorems

10.1 Theorem 3 (S3)

$$\tilde{S} = \bar{S} + \kappa(S - \bar{S}) = \bar{S} + \kappa(Q + \eta N - \bar{Q} - \eta \bar{N}). \quad (43)$$

$$\frac{\partial \tilde{S}}{\partial Q} = \kappa'(S - \bar{S}) = \frac{\partial \tilde{S}}{\partial(\eta N)}. \quad (44)$$

κ -

Theorem 3 (Noise-Difficulty Indistinguishability) is the logical foundation of the κ Suppression Paradox. Since no auditor observing only CERCIS scores can distinguish label noise from intrinsic difficulty, any suppression strategy based solely on score magnitude (which is exactly what κ does) cannot selectively suppress noise without harming signal. This implies **zero noise-signal selectivity for any κ suppression**.

10.2 Spring SE-1

$$\mathbb{E}[\|\theta_t - \theta^*\|^2] = O\left(\frac{1}{\kappa_F \cdot t}\right) = O\left(\frac{1}{(1 - c_2\lambda)t}\right), \quad (45)$$

$\lambda \rightarrow \lambda_c$ Spring

Spring SE-1’s self-evolving gating relies on CERCIS score gradient signals for Robbins-Monro stochastic approximation. κ suppression attenuates the effective gradient by a factor of κ_F , degrading the convergence rate as shown above, diverging near $\lambda \rightarrow \lambda_c$ and signaling the collapse of Spring dynamics.

11 Numerical Experiments

11.1 Experimental Design

- 1** : $S_i = \mu + \varepsilon_i \varepsilon_i \sim (0, \sigma^2)\mu = 0\kappa$
- 2** -: $S_i = \mu_i + \varepsilon_i \mu_i \sim (0, \sigma_\mu^2)\varepsilon_i \sim (0, \sigma_\varepsilon^2)\lambda_c$
- 3** : $N = 20M = 50010$ “” 10 “”

Three experimental classes are designed to validate theoretical predictions (see `verify_kappa.py`):

(1) Pure Gaussian noise — verifying non-selective compression; (2) Mixed signal-noise — verifying phase transition at λ_c and elite erosion; (3) Realistic audit simulation — with $N = 20$ experts on $M = 500$ data points, 10 “elite” and 10 “ordinary”, verifying elite suppression ratio and diversity decay.

11.2 Key Numerical Results

- (i) λ_c : $2\lambda = \lambda_c \pm 0.05\lambda_c p < 0.001$ bootstrap
- (ii) : $3R_{\text{elite}} = 0.62 \pm 0.03$ vs $R_{\text{non-elite}} = 0.78 \pm 0.02\lambda = \lambda_c 62\% 78\% t\text{-test } p < 10^{-6}$
- (iii) : $3KLt = 1038\% \pm 5\% e^{-2\lambda_c \bar{\sigma}^2 \cdot 10} \approx 0.41\chi^2 p = 0.23$
- (iv) : Protocol `??t` $\approx 15\lambda\lambda_{\text{max}} = 5\lambda_c\lambda^* \approx 0.8\lambda_c Q_\kappa = 0.91Q_0\lambda = \lambda_c 0.76Q_0$

Numerical experiments confirm: (i) λ_c accuracy — slope change at $\lambda_c \pm 0.05\lambda_c$ ($p < 0.001$); (ii) elite suppression ratio — $R_{\text{elite}} = 0.62$ vs $R_{\text{non-elite}} = 0.78$ ($p < 10^{-6}$); (iii) diversity decay — KL divergence decays to 38% of initial at $t = 10$ (theory predicts $\sim 41\%$, $p = 0.23$); (iv) adaptive annealing restores Q_κ to $0.91Q_0$, outperforming fixed $\lambda = \lambda_c$ at $0.76Q_0$.

12 Discussion: Cultural and Institutional Roots of Suppression

(1) : “”——SCX

(2) : “”——“”

(3) : L1/L2dropoutgradient clipping κ

Despite the mathematical self-defeat of κ suppression proved here, it is pervasive in practice. This is not accidental — suppression satisfies three deep psychological and social needs: (1) the illusion of surface fairness through apparent consensus; (2) institutional incentives for risk aversion in hierarchical organizations; (3) path dependency from ML regularization techniques, where practitioners uncritically generalize suppression from training to evaluation, ignoring that suppression in training versus target evaluation serves fundamentally different purposes.

12.1 Open Problems

OP1 κ : Lipschitz $\kappa\eta\kappa(\delta) = \delta$

OP2 κ : κNash

OP3 κ : AR-Theorem $A\kappa\text{AR}\kappa$

OP4 SCX-native: Cercis Score“” $\eta\kappa$

13 Conclusion

(1) $\kappa\kappa_I\kappa_F\kappa_B$

(2) κ

(3) $\lambda_c = \eta/(2\sigma_\varepsilon(1 + \eta))$

(4) κ -

: ηCercis

This paper rigorously establishes the mathematical foundation of the κ Suppression Paradox within the SCX framework. Core findings include: (1) the triple mechanism attacking cognition, correction, and learning; (2) the Elite Erosion Theorem proving disproportionate penalty on high-quality experts; (3) the Suppression Phase Transition Theorem with critical threshold λ_c ; (4) the adaptive κ -annealing protocol. **Engineering takeaway:** If your audit system uses any form of score suppression, you are systematically destroying the value of your best experts. Stop suppressing. Adopt adaptive η tuning or the Cercis consistency condition instead.

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/ **Verification Script:** See accompanying file `verify_kappa.py` for complete numerical validation of all theoretical claims in this paper.